EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2018
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME								
CIVICS GROUP	1	7	-			REGISTRATION NUMBER		

H2 Biology

9744/02

Paper 2 Structured Questions

14 September 2018

2 hours

Candidates are to **answer all questions** in this question booklet.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
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10	
Total	**100**

This document consists of **29** printed pages and **1** blank page.

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Section A

Answer **all** the questions in this section.

1 The synthesis of collagen is shown in Fig. 1.1.

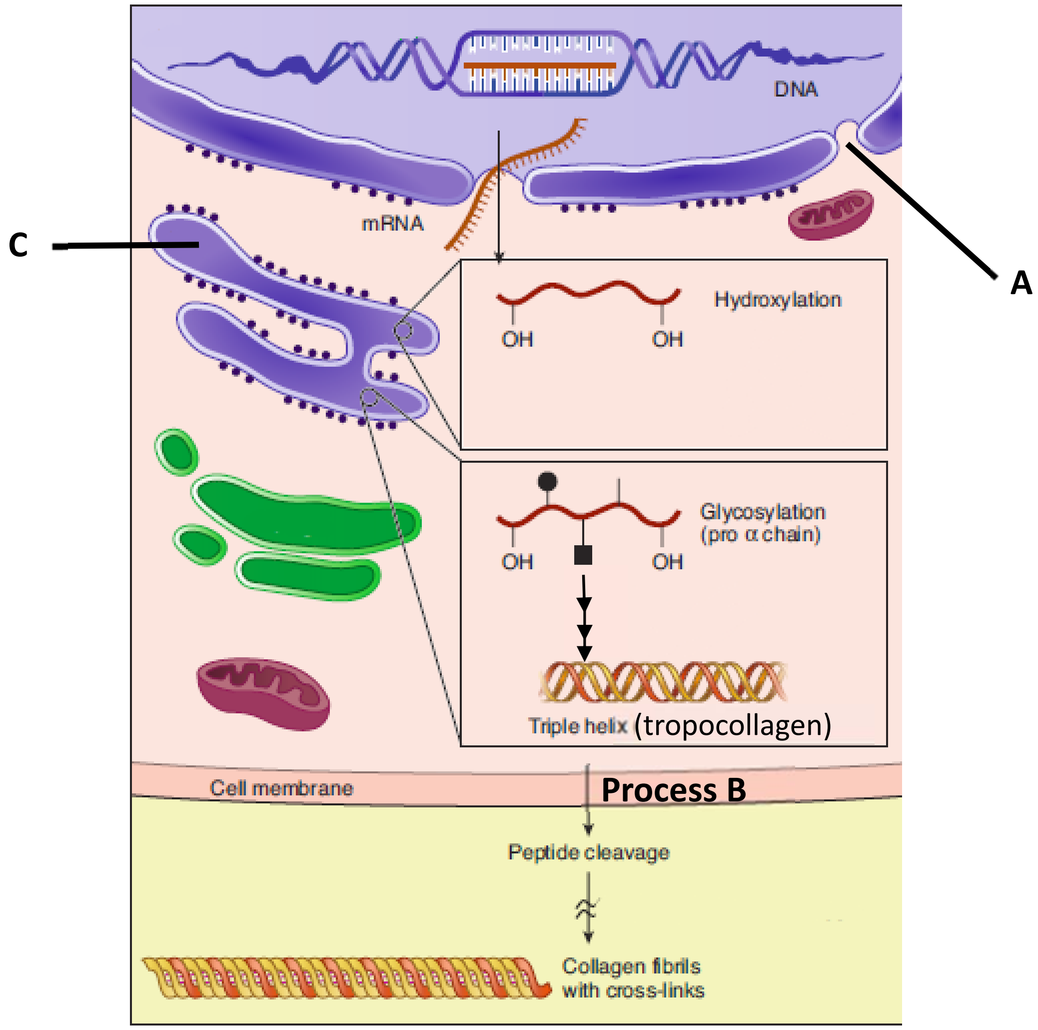

Fig. 1.1

(a) **(i)** Describe two important functions of structure **A** in the synthesis of collagen.

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....[2]

Structure A is the nuclear pore;

1. **Allows mature mRNA to <u>leave nucleus</u> and <u>enter cytosol</u> for <u>translation into collagen</u> by ribosome;**
2. **Allows <u>RNA polymerase</u>/ <u>ribonucleotides</u> to <u>enter nucleus</u> to synthesise <u>mRNA/tRNA</u> during <u>transcription</u> of <u>collagen gene</u>;**
3. **Allows <u>exit of mRNA to ribosomes</u> to be translated;**

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4. **Allows tRNA to leave nucleus for amino acid activation/ amino acid attachment for use in translation during collagen synthesis;**
(any 2)

(ii) Tropocollagen leaves the cell to be assembled to form collagen fibrils via **Process B**. Outline **Process B**.

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1. **Process B is exocytosis;**
2. **Secretory vesicle containing tropocollagen is transported to cell surface membrane and fuses with the cell surface membrane to release tropocollagen extracellularly;**
3. **This process requires ATP;**

(b) Suggest how chemical modification such as hydroxylation in organelle **C** results in collagen having a high tensile strength.

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1. **Hydroxylation is the process where hydroxyl groups are added/ attached/ joined to the polypeptide chain;**
2. **To enable formation of numerous interchain hydrogen bonds during the formation of tropocollagen triple helix, giving rise to high tensile strength;**

Fig. 1.2 shows two electron micrographs. One of the electron micrograph shows part of a normal cell while the other electron micrograph shows part of a cancer cell. The white arrows point to an organelle within the cell. The appearance of this organelle in both cell types were visibly different. The cancer cell had a higher activity than the normal cell.

Organelle (indicated by white arrows) in a normal cell

Organelle (indicated by white arrows) in a cancer cell

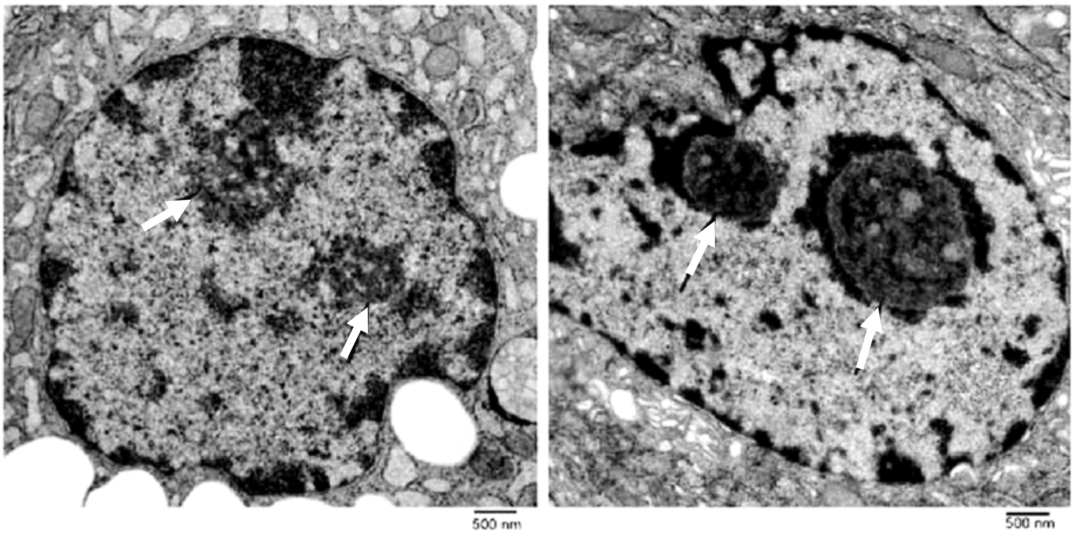

Fig. 1.2

(c) **(i)** Describe the visible difference between the organelle indicated by the white arrows in Fig. 1.2, between the cancer cell and the normal cell.

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1. **Nucleoli in the cancer cell were <u>denser/darker/more darkly stained</u> and <u>larger</u>;**

(ii) Account for the higher activity of the cancer cell.

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1. **Nucleoli are the <u>sites for rRNA synthesis</u> which are key components of <u>ribosomes</u> required for <u>translation</u>;**
2. **Larger and denser nucleoli in cancer cells due to <u>increased rRNA</u> enable cancer cell to achieve <u>higher rates of protein synthesis</u> to enable <u>excessive proliferation/growth</u>;**

[Total: 10]

2 The rate of glycolysis is regulated by the action of ATP on the enzyme phosphofructokinase (PFK), the third enzyme in the glycolysis pathway. PFK catalyzes the formation of fructose 1,6-bisphosphate from fructose-6-phosphate and ATP.

A graph of PFK against F6P concentration would exhibit the 'sigmoidal curve' typical of allosteric enzymes.

Fig. 2.1 shows the effect of substrate F6P (fructose 6-phosphate) concentration on the activity of PFK at different ATP concentrations. PFK is an allosteric enzyme with a quaternary structure that is inhibited by high levels of ATP. High levels of ADP and AMP will increase activity of the enzyme.

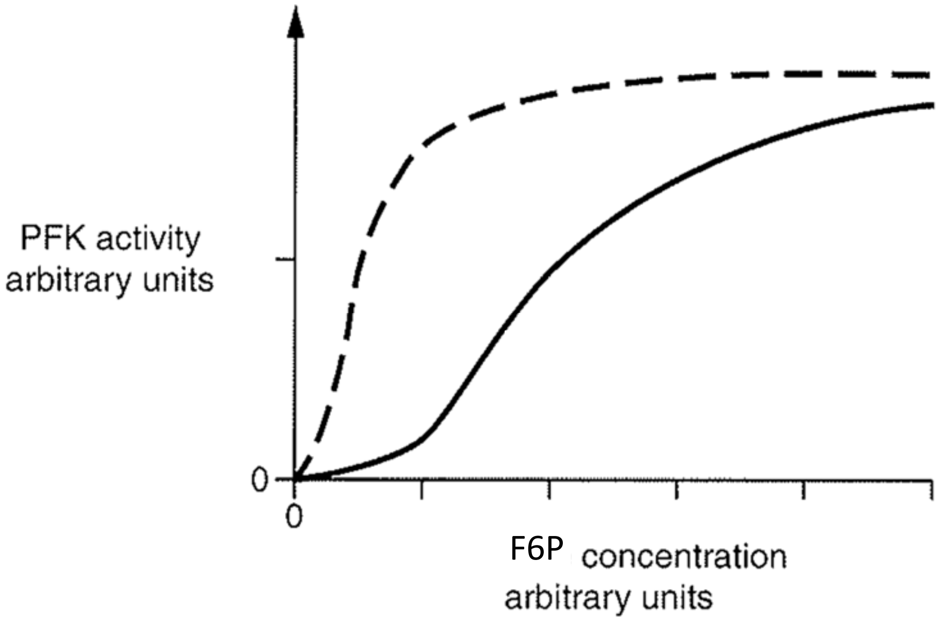

Fig. 2.1

(a) **(i)** With reference to PFK, explain the term '*allosteric enzyme*'.

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5. PFK is an enzyme that has <u>more than 1 subunits/ multimeric/ 4 subunits</u>
6. with the <u>allosteric site</u> where <u>AMP/ADP/ATP</u> can bind to;

Reject references to 'site other than active site'

(ii) On Fig. 2.1, label the graphs which reflect low ATP and high ATP concentrations respectively. [1]

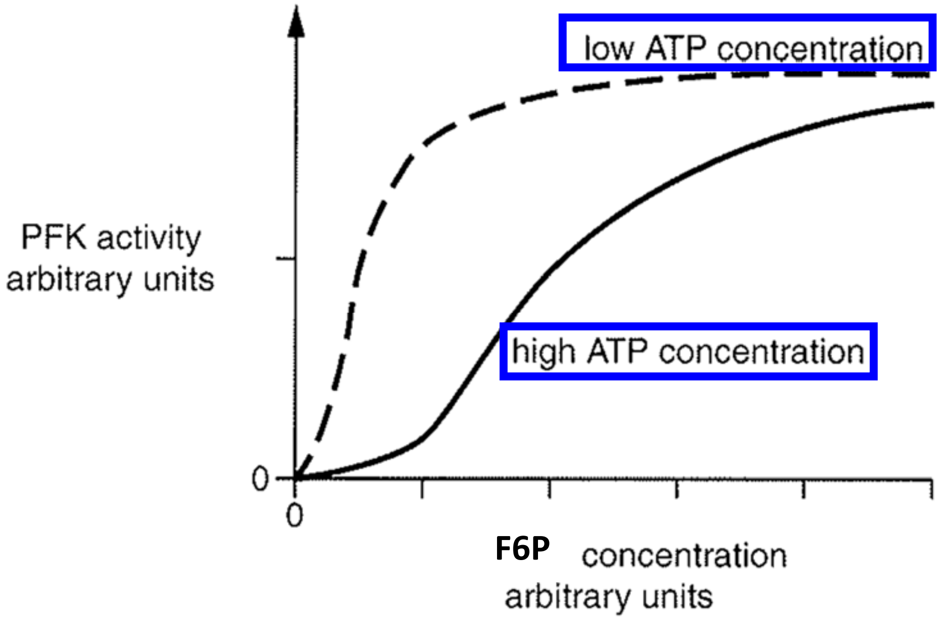

Fig.2.1

All or none;

(b) Enzyme researchers have used a model of allosteric enzyme mechanism called the 'symmetry model' to explain the action of PFK. F6P binds with great affinity only to the active state but not to the inactive state. Binding of one F6P molecule will progressively shift PFK structure from the inactive state to the active state.

A graph of PFK against F6P concentration would exhibit the 'sigmoidal curve' typical of allosteric enzymes.

(i) Explain how the binding of a molecule like AMP to PFK can increase the activity of PFK.

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....[2]

1. **Binding of AMP to allosteric site of PFK will induce <u>conformation change</u> in PFK from <u>inactive to active state</u>;**
2. **Conformation change will help to <u>stabilize the active state</u> of PFK / and will lead to the <u>active site</u> being made <u>more complementary</u> to F6P;**

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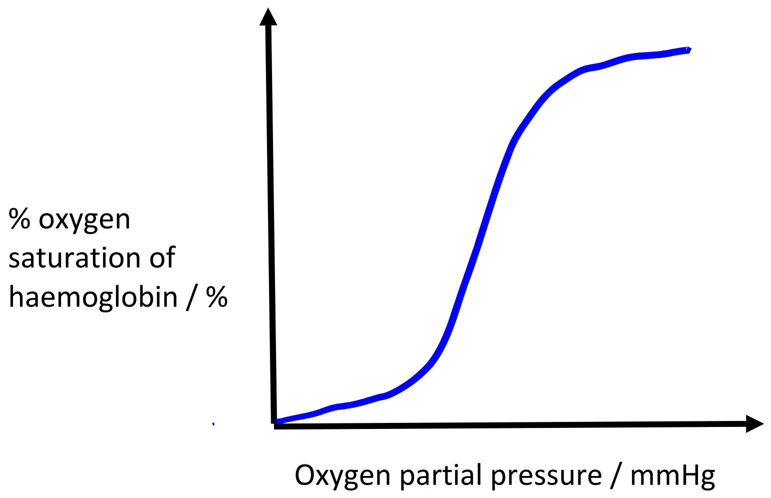
(ii) The enzymatic mechanism in PFK is similar to that of the oxygen-binding mechanism of the transport protein, haemoglobin. Sketch the shape of the oxygen binding graph for haemoglobin at increasing concentrations of oxygen in the space below. [1]



S-shaped curve starting from origin (gentle slope followed by steep curve, followed by gentle slope)

(iii) Using your knowledge of the structure of haemoglobin, explain the shape of the graph you drew in **(b) (ii)**.

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1. **Cooperative binding of oxygen - Binding of one oxygen molecule to one haemoglobin subunit induces a structural / conformational change in the other 3 subunits;**

2. **This leads to an increase in their affinity for oxygen, thus lead to rapid loading of the other 3 oxygen molecules / steep increase in % oxygen saturation of haemoglobin;**

3. **Gentle slope/plateau of graph at the end reflects saturation of the haemoglobin binding sites in the haemoglobin molecules present;**

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(c) Explain the significance of regulation of PFK by ATP to AMP ratios.

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1. **High glycolysis rates lead to high respiratory rates, producing high levels of ATP, <u>therefore high ATP:AMP</u> can <u>inhibit PFK</u> to <u>slow glycolysis</u> because there is <u>sufficient</u> levels of ATP in the cell;**
2. **till these ATP are <u>depleted to low ATP:AMP levels</u> where <u>PFK is activated to increase glycolysis/</u>respiration rate to <u>increase levels of ATP</u>;**
3. **This reduces the need for glycolysis to be at constantly high rates, leading to <u>efficient use of resources in the cell</u> (e.g. ADP, ATP, glucose and pyruvate etc).;**

(Any 2)

3 Bone marrow contains stem cells that divide by mitosis to form blood cells. The fate of a stem cell was tracked and it was recorded that during the observed duration the stem cell divided asymmetrically each time.

Fig. 3.1 shows changes in the mass of DNA in a human stem cell from bone marrow during three cell cycles.

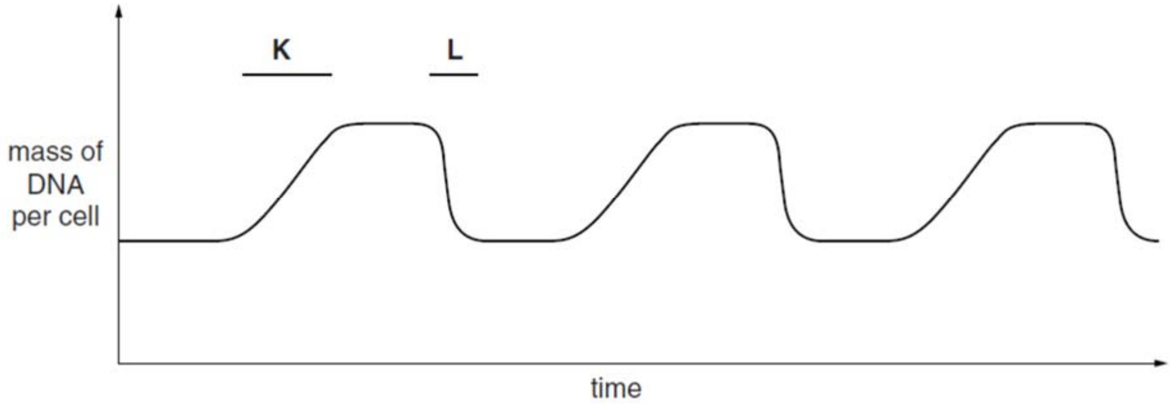

Fig. 3.1

(a) With reference to the information provided above,

(i) Describe what happens to bring about the changes in the mass of DNA per cell at time period **K** and at time period **L**.

K ....

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L ....

....[2]

1. **K: <u>DNA replication/ synthesis</u> during <u>S phase of interphase;</u>**
2. **L : <u>cytoplasmic division</u> during <u>cytokinesis;</u>**

(ii) State one function of these stem cells undergoing the above type of cell division.

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....[1]

1. **To <u>replace blood cells</u> that <u>die</u> due to <u>injury/ wear and tear/ disease;</u>**

(iii) The process of meiosis is significant to natural selection in evolution.

Explain this significance.

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For Examiner's Use

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1. Meiosis gives rise to genetically different haploid gametes;
2. During Prophase I of meiosis, crossing over between chromatids of homologous chromosomes would give rise to different combinations of alleles

   OR

   During Metaphase I and Anaphase I, independent assortment and separation of homologous chromosomes would give rise to different combinations of maternal and paternal chromosomes;

3. Resulting in genetic variation in diploid organisms leading to different phenotypes/ variation in phenotypes;

4. those with advantageous phenotypes are selected for, resulting in change in allele frequency in the population over time;

   (1 and 2 + 3 / 4 OR 1+ 3 and 4)

A bone marrow cell was extracted and observed under the electron micrograph shown in Fig. 3.2. The student focused on an organelle which he described as having "an envelope surrounding genetic material containing both darker and lighter stained patches, distinct from the site where ribosomal subunits were assembled".

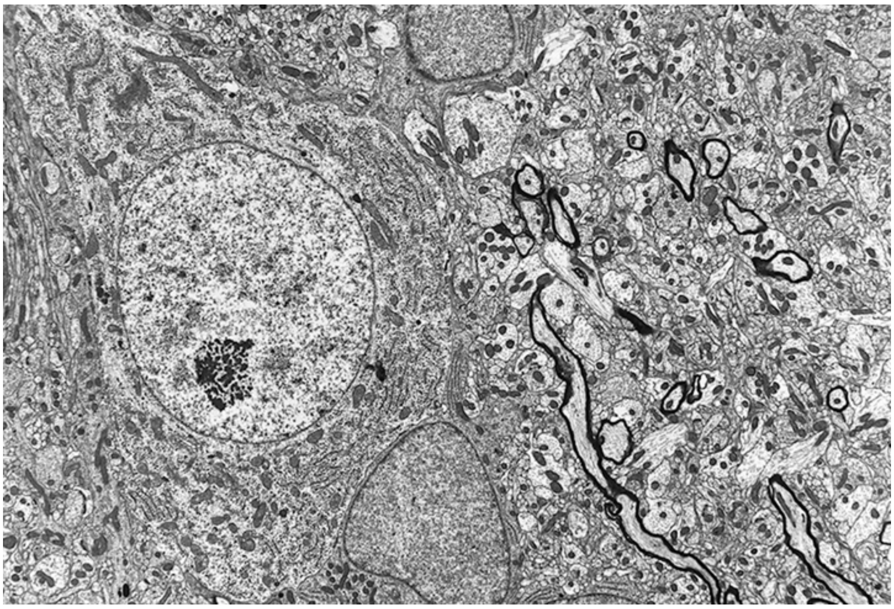

Fig. 3.2

(b) Explain the significance of the "darker and lighter stained patches" that the student referred to, in a cell undergoing differentiation.

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1. **The darker stained patches are heterochromatin while the lighter stained patches are euchromatin;**
2. **The genes that are found within heterochromatin are transcriptionally inactive/not expressed while the genes found in euchromatin are transcriptionally active/are expressed;**
3. **Thus producing specific proteins that allow the differentiated cell to perform specific functions;**

The use of embryonic stem cells (ESCs) for stem cell therapy and research is controversial and considered by many people as unethical. Scientists have circumvented this issue through the use of induced pluripotent stem cells (iPSCs) as an alternative to ESCs.

Fig. 3.3 summarises the procedure for obtaining iPSCs and its use.

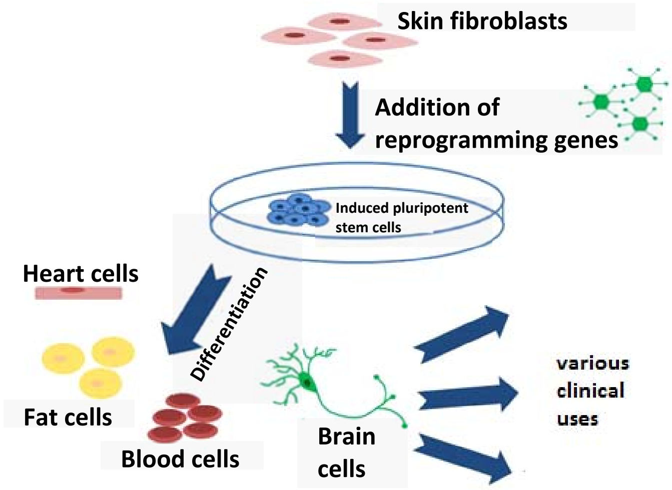

Fig. 3.3

(c) Explain why the use of iPSCs is preferred over ESCs.

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1. **Since iPSCs can be derived directly from adult tissues/specialised somatic cell/ skin cells, it does not destroy any human embryos unlike ESCs;**
2. **iPSCs from adult/specialised somatic cell/ skin cell can be easily obtained without risk to the donor, whereas obtaining the embryo to isolate ESCs is more invasive;**
   (R: easy vs difficult because no elaboration)
3. **In contrast to ES cells extracted from human embryos, iPSCs derived from a patient's own cells would open the possibility of generating patient-specific cells, which will not be rejected by the immune system upon transplantation; (A: idea of iPSCs obtained**

from self vs embryos not from self thus contain antigens that will be rejected by immune system);

4. **Sources of obtaining iPSCs is more easily accessible compared to the source of obtaining ESCs/ obtaining the embryo;**
(any one, answer must make comparison with ESCs)

[Total: 10]

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4 Fig. 4.1 shows part of a DNA molecule.

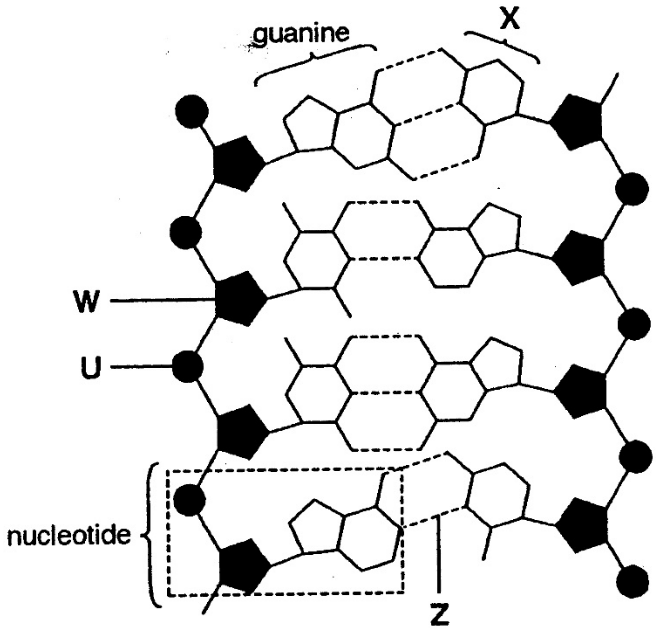

Fig. 4.1

(a) (i) Name **W** to **Z**. [2]

W ……………………………………………………….

U ……………………………………………………….

X ……………………………………………………….

Z ……………………………………………………….

1. **W = Deoxyribose (R: sugar, ribose, pentose)**
2. **U = Phosphate group (R: phosphate head)**
3. **X = Cytosine (R: nitrogenous base, pyrimidine)**
4. **Z = Hydrogen bond**

4 correct = 2 marks, 2-3 correct = 1 mark, less than 2 correct = 0 marks

(ii) Give **one** advantage of DNA having two strands.

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1. **Collective/numerous hydrogen bonds between the complementary base pairs stabilise the double helical structure/molecule;**
2. **One strand act as a template for repair during DNA replication;**
3. **Protects / shields the hydrophobic nitrogenous bases from the hydrophilic medium;**

(b) Fig. 4.2 shows a linear chromosome undergoing the first round of DNA replication.

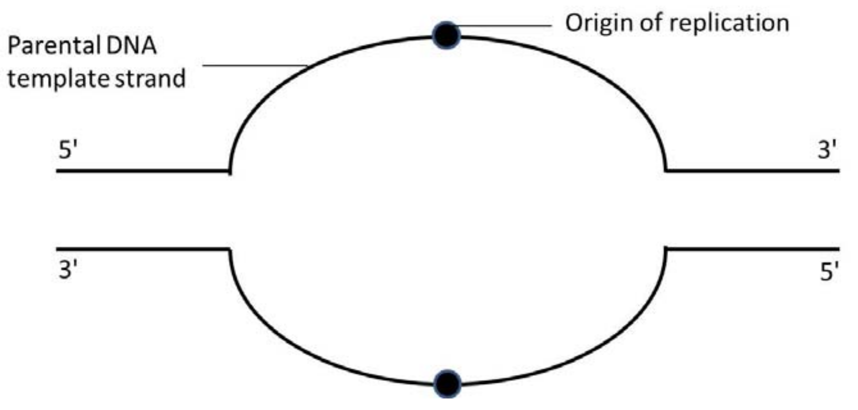
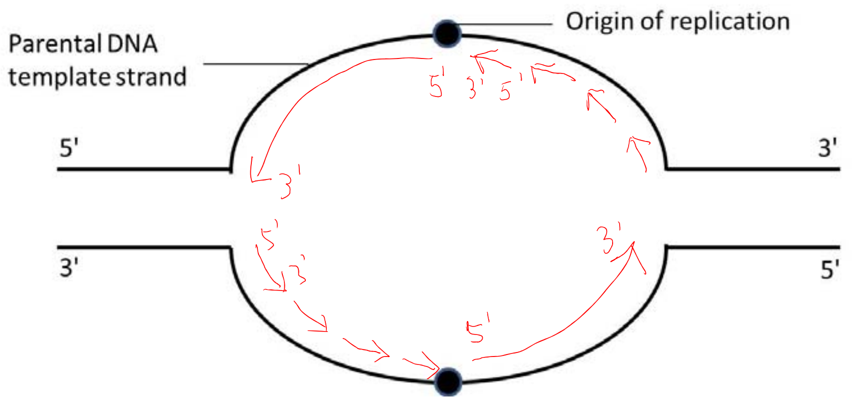


Fig. 4.2

(i) On Fig. 4.2, draw the direction of DNA synthesis for the leading strands using solid arrows (→) and for the lagging strands using dashed arrows (- - - >).

[1]



1. **All directions of leading and lagging strands (Okazaki fragments) correctly indicated;**
2. **Lagging strands must show several Okazaki fragments in correct directions;**

Points 1 and 2 must be present for full credit. Not marking for 5'/ 3' labels, but arrow head needed.

(ii) State **two** ways DNA replication is different from transcription.

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DNA Replication	Transcription
DNA polymerase used to polymerise nucleotides/synthesize new strands;	RNA polymerase used to polymerise nucleotides/synthesize new strands
Helicase used to unzip DNA strands;	RNA polymerase unzip DNA strands
Both parental strands used as templates;	Only one strand used as template
DNA strands produced;	mRNA/tRNA/rRNA strand produced
Deoyribonucleotide monomers;	Ribonucleotide monomers
Daughter DNA molecule double-stranded;	mRNA molecule single stranded
AVP;	AVP

(c) Fig. 4.3 is an electronmicrograph showing the transcription of genes for ribosomal RNAs at adjacent positions along the chromosome in a eukaryotic cell.

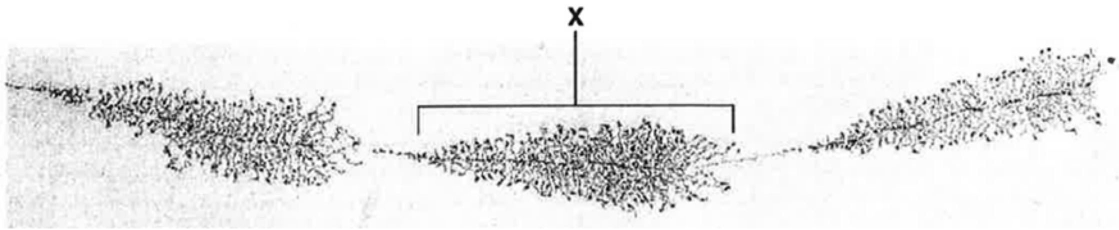

Fig. 4.3

(i) Describe and explain the pattern of transcription visible on the part of the DNA coding for ribosomal RNA, labelled **X** in Fig. 4.3.

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1. The pattern shows shorter strands of rRNA emerging on the left and with longer strands of rRNA seen on the right;

2. This is because RNA polymerase transcribes DNA from 3' to 5' direction/ synthesises rRNA from 5' to 3' direction, hence the rRNA strands on the right (near 5' end of DNA template) were the longest (transcription started earlier); (ora)

(ii) Suggest possible roles for the non-transcribed DNA that is found between the ribosomal genes.

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[Total:10]

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1. **The non-transcribed DNA may be enhancer sequences which are bound by activators which increase the transcription rate;**

2. **The non-transcribed DNA may be silencer sequences which are bound by repressors which decrease the transcription rate;**

3. **The non-transcribed DNA may be promoter sequences which are bound by general transcription factors and RNA polymerase bind to form transcription initiation complex to start transcription;**

5 Human immunodeficiency virus (HIV) causes the condition known as acquired immunodeficiency syndrome (AIDS) in humans.

The development of AIDS in HIV infected persons may take place many years after infection with the virus, e.g. 8.5 years later, as illustrated in Fig. 5.1.

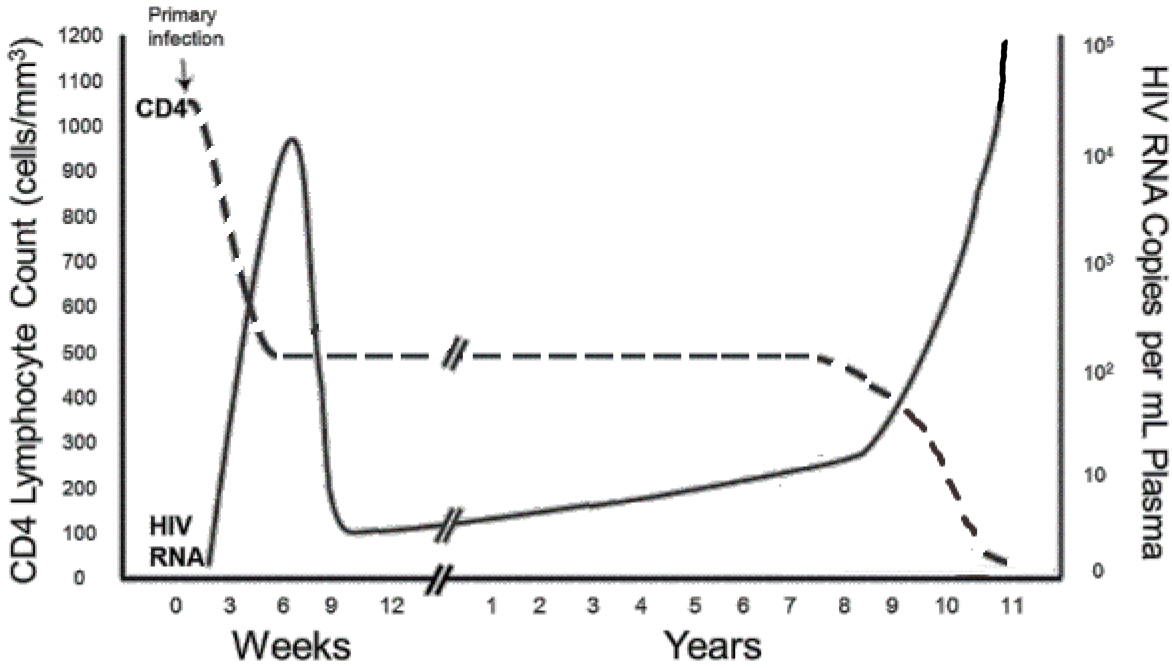

Fig. 5.1

(a) With reference to Fig. 5.1,

(i) Explain the level of HIV RNA detected in the plasma from week 9 onwards after infection.

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....[3]

1. **From week 9 to 8.5 years after infection**, level of HIV RNA in plasma was **low**, at **10 and below per ml of plasma**. **8.5 years after infection**, there is a **steep rise** in amount of HIV RNA to **more than 10^5 per ml plasma**;
2. Low level of HIV RNA is due to **latent period/ no replication/ provirus not expressed/ provirus not transcribed to viral RNA** of HIV virus;
3. Followed by **activation** of **provirus** whereby provirus is **transcribed / expressed** to form HIV RNA;

(ii) Explain why AIDS patients have compromised immune systems.

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1. HIV **infects** **T helper cells** and causes **lysis of T helper cells** when they are released from these host cells;
2. This results in a **significant drop** in amount of T helper cells in plasma to **almost zero per ml plasma**;
3. Without cytokines from T helper cells, naïve **B cells are not activated** / **no plasma cells form** and **no antibodies** are produced;
4. Without cytokines from T helper cells, **macrophages** also **not activated/ does not carry out phagocytosis efficiently** for destruction of pathogens;

(1 + 2 and 3/4)

(iii) With a weakened immune system, AIDS patients are susceptible to opportunistic infections, whereby infections occur more frequently and are more severe.

On Fig. 5.1, indicate clearly with an arrow along the X axis when HIV infected patients succumb to opportunistic infections. [1] **(accept from 9 years onwards to 11)**

(b) Scientists are trying to develop an effective vaccine to protect people against HIV. However, there are three main problems: HIV causes immune system of infected individuals to be compromised, HIV rapidly enters host cells and HIV has a high mutation rate.

(i) State one structural feature of HIV that accounts for its high mutation rate and explain why this means that a vaccine might not be effective against HIV.

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1. **Reverse transcriptase** **no proofreading ability**;
   OR
   HIV genome is **single stranded RNA**, therefore **no backup strand** **for repair** of new strand;

2. High mutation rate results in many different strains of HIV with **different antigens** displayed / **antigen** on HIV **changes;**
3. **Antibody** formed by memory cells from vaccination **cannot bind** to new antigen to **neutralise virus / to prevent virus from binding to host cell;**
   OR
   only bind to **antigens of the particular strain** the vaccine was based on;

(ii) State the type of immunity conferred by vaccines.

…………………………………………………………………………………………………[1]

Artificial active immunity;

[Total: 10]

6 A plant breeder crossed a plant from a pure-bred line of tomato plants with red uniformly pigmented fruit with a plant from a pure-bred line of plants with orange-coloured fruit, but with unattractive dark patches at the base of the unripe fruit. The resulting generation all produced red fruit with dark patches. Plants from this generation were interbred. The resulting progeny showed the following numbers of plants in each of the three phenotypes:

Red fruit with dark patches	98
Red uniformly pigmented fruit	46
Orange fruit with dark patches	44

(a) Explain how genotype is linked to the phenotype.

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……[2]

1. **The genotype of an organism is its <u>genetic makeup</u> which is the <u>combination of alleles</u> situated on <u>corresponding loci</u> on <u>homologous chromosomes</u>;**
2. **Alleles code for <u>proteins</u> which will make up the <u>phenotype</u> of the organism;**

(b) The genes involved in the cross above are completely linked. Explain the meaning of the term 'complete linkage'.

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……[2]

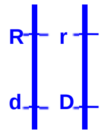
1. **Both genes involved in the cross are adjacent to each other on the same chromosomes;**
2. **No crossing over/ No formation of recombinant gametes / always inherited together;**

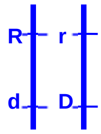
(c) Draw a genetic diagram to show how the second cross could lead to the three phenotypes.

Symbols: Let R be the dominant allele for red fruit and r be the recessive allele for orange fruit.
Let D be the dominant allele for dark patched skin and d be the recessive allele for pigmented skin;

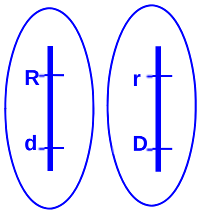
Parental Phenotypes | **Red Fruit, Dark Patches** **x** **Red Fruit, Dark Patches**

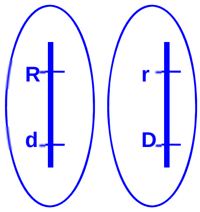
Parental Genotypes;



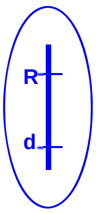
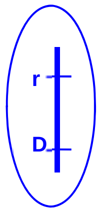
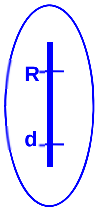
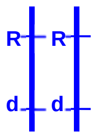
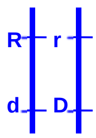
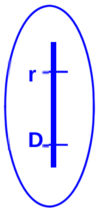
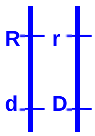
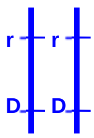


Meiosis
Gametes (2 sets);





Fertilisation (Punnett Square showing correct genotypes and phenotypes);

		
	 **Red fruit, pigmented skin**	 **Red fruit, dark patches**
	 **Red fruit, dark patches**	 **Orange fruit, dark patches**

Offspring Genotypic ratio
red fruit, dark patches : red fruit, pigmented skin : orange fruit, dark patches

Offspring Phenotypic ratio;
2 : 1 : 1

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[5]

The χ^2 distribution table (Table 6.1) and equation to calculate χ^2 is shown below. Using the formula, the calculated χ^2 value for the cross was 0.38.

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

Table 6.1

Degree of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(d) Using the calculated value of χ^2 and the table of probabilities provided in Table 6.1, explain what conclusion can be drawn from the data.

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1. **<u>df = 3-1 = 2</u>, p>0.10, since <u>p>0.05</u>, at a level of significance of 5% we <u>do not reject the null hypothesis</u> that states that the expected results and observed results are not significantly different;**
2. **Thus, there is <u>no significant difference</u> between the <u>expected and observed results</u> and <u>any difference is due to random chance alone</u>;**

[Total: 11]

7 Fig. 7.1 shows a CREB (cAMP response element binding) signalling pathway. This pathway is triggered when ligand X binds to a receptor at the cell membrane.

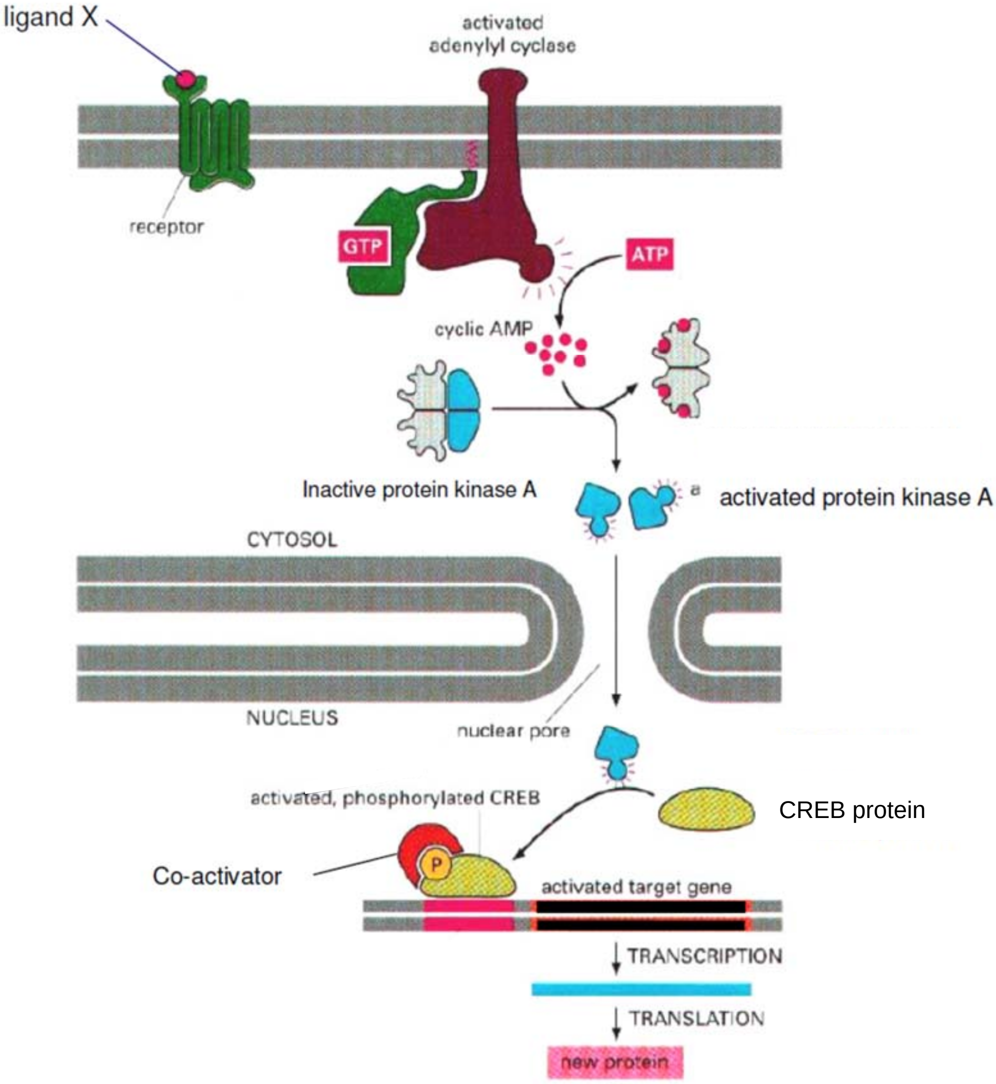

Fig. 7.1

(a) With reference to Fig. 7.1,

(i) Describe precisely the structure of the G protein coupled receptor and explain how the described structure enables it to perform its function.

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1. **GPCR is a receptor protein with <u>seven transmembrane domains</u>, consisting of <u>hydrophobic regions</u> forming <u>hydrophobic interactions</u> with the <u>hydrophobic fatty acid tails/nhydrophobic core of the phospholipid bilayer</u>;**
2. **Allows GPCR to be <u>embedded</u> on the <u>cell surface membrane</u> to allow it <u>bind/ interact</u> with <u>hydrophilic/ polar/ large/ charged ligand</u>;**

OR

3. **GPCR has an extracellular ligand-binding site that is complementary in shape and charge to the ligand allowing ligand to bind**
4. **to induce conformational change in intracellular domain of receptor leading to signal transduction;**

OR

5. **GPCR has an intracellular domain that associates with a G-protein;**
6. **Change in conformation of intracellular domain of receptor leading to signal transduction;**

(ii) Describe how protein kinase A (PKA) is activated.

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1. **cAMP binds to 2 subunits of inactive protein kinase A;**
2. **the remaining 2 subunits dissociate and become activated protein kinase A;**

(iii) State the role of PKA in this signaling pathway?

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....[1]

1. **To phosphorylate the CREB protein so that co-activator can bind to it.**

(b) CREB is an activator protein that binds to its specific control element to upregulate the transcription of the target gene.

Explain how an activator protein like CREB can upregulate transcription of a gene.

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1. **CREB will <u>bind</u> to its <u>enhancer</u> on DNA;**
2. **causing the <u>spacer DNA</u> between the promoter of the gene and enhancer to <u>bend</u>**
   OR
   Recruit <u>histone acetylase</u> / <u>chromatin remodelling complex</u> to <u>decondense chromatin</u>;
3. **Thus promoting the <u>assembly of RNA polymerase</u> and <u>general transcription factors</u> at the <u>promoter</u> to form <u>transcription initiation complex</u>;**

(c) CREB is known to be involved in other signaling pathways. With reference to Fig. 7.1 or otherwise, suggest how it is possible for CREB to bind to other control elements so as to regulate the transcription of different genes.

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1. **CREB protein can be <u>phosphorylated at different sites</u> by <u>PKA</u>, leading to <u>different conformation changes</u> that allow CREB to <u>bind</u> to <u>different control elements</u>;**

2. **Different co-activators may be recruited and bind to CREB, leading to <u>different conformation changes</u> that allow CREB to <u>bind</u> to <u>different control elements;</u>**

(any one)

[Total: 9]

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8 A student investigated the effect of different wavelengths of light on the rate of photosynthesis. She used the apparatus shown in Fig. 8.1 below.

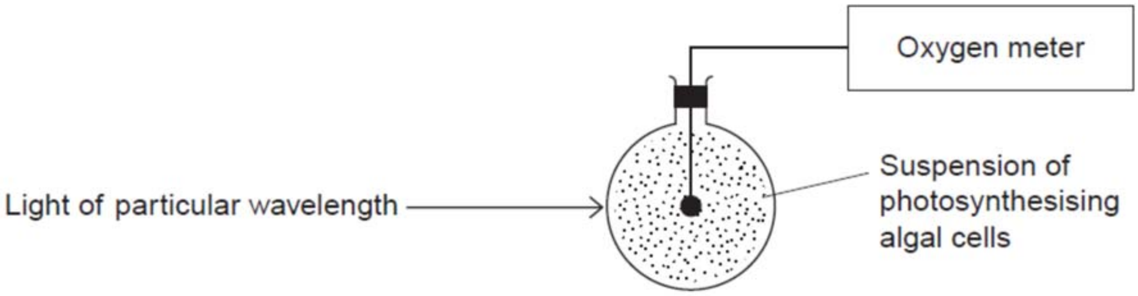

Fig. 8.1

(a) State a condition that is to be kept optimum in the investigation shown in Fig. 8.1.

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....[1]

1. <u>Carbon dioxide concentration/ temperature/ enzyme concentration/ algae concentration;</u>

(b) The student did not use a buffer to maintain the pH of the solution and observed that the pH of the solution changed over the course of the investigation.

Predict and explain the effect of this pH change on photosynthesis in the investigation.

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1. **pH <u>increases</u>;**
2. **this <u>decreases</u> the rate of photosynthesis as the pH is <u>no longer at optimum</u> for the enzymes required for photosynthesis + <u>1 e.g. of enzyme</u> (rubisco/ NADP* reductase);**

The student plotted her results below in Fig. 8.2.

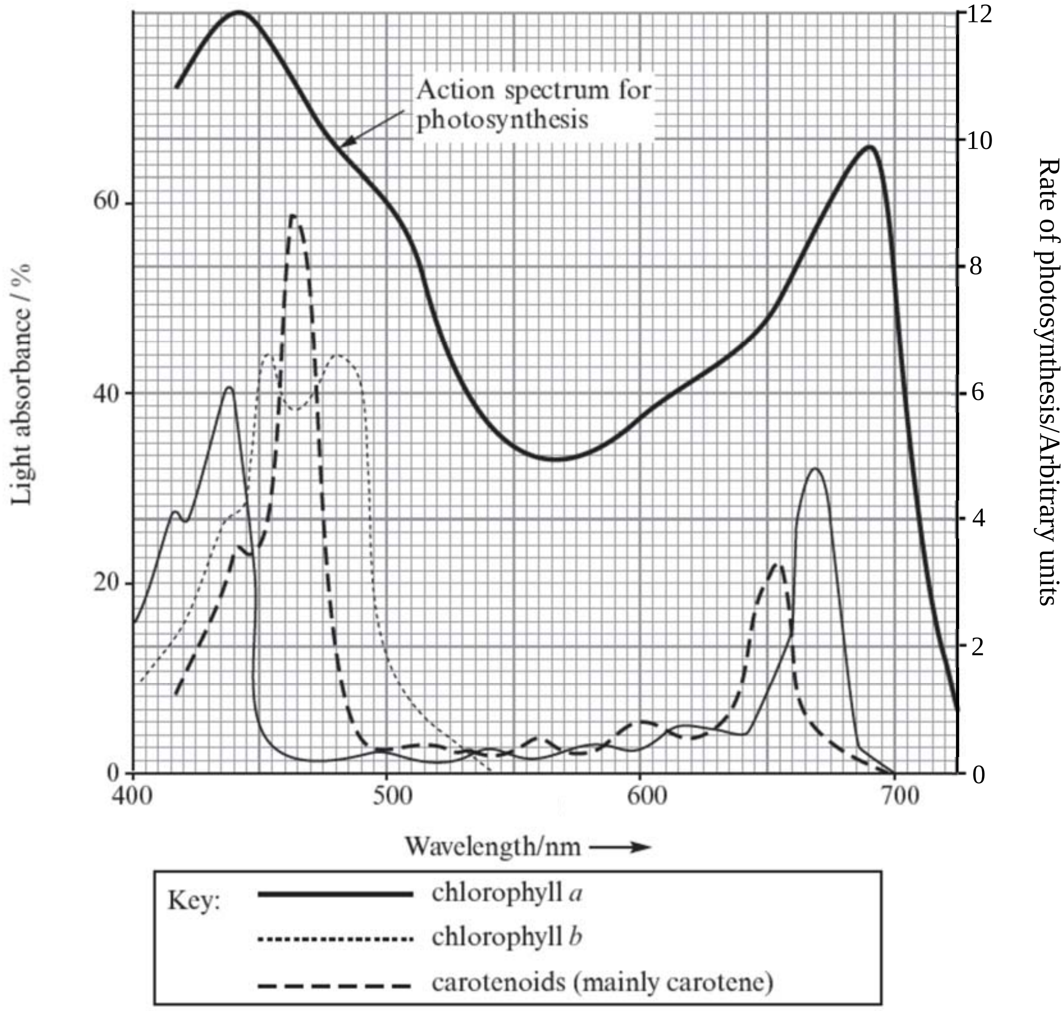

Fig. 8.2

(c) Chlorophyll *a* functions as a primary photosynthetic pigment. With reference to Fig. 8.2, explain the advantage of a plant having accessory pigments like carotenoids.

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1. **Increases the range of wavelengths that can be absorbed such as 465nm/ 655nm;**
2. **To increase photosynthesis rate so that the plant cell to produce more glucose for energy storage as starch/ for growth as cellulose/ for respiration/ ATP synthesis;**

   OR

   To play a photoprotective role so as to absorb and dissipate excess light energy that could damage chlorophyll;

(d) Explain the relationship between the absorption spectrum and the action spectrum.

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1. **The shape of action spectrum correlates/is similar to/ follows a similar pattern with that of absorption spectrum;**
2. **This shows that different wavelengths of light that are absorbed are used for photosynthesis;**

One of the most significant consequences of climate change is global warming. Fig. 8.3 shows the impact of global warming on crop yield on two temperate cereal crops, corn and soybean in Africa.

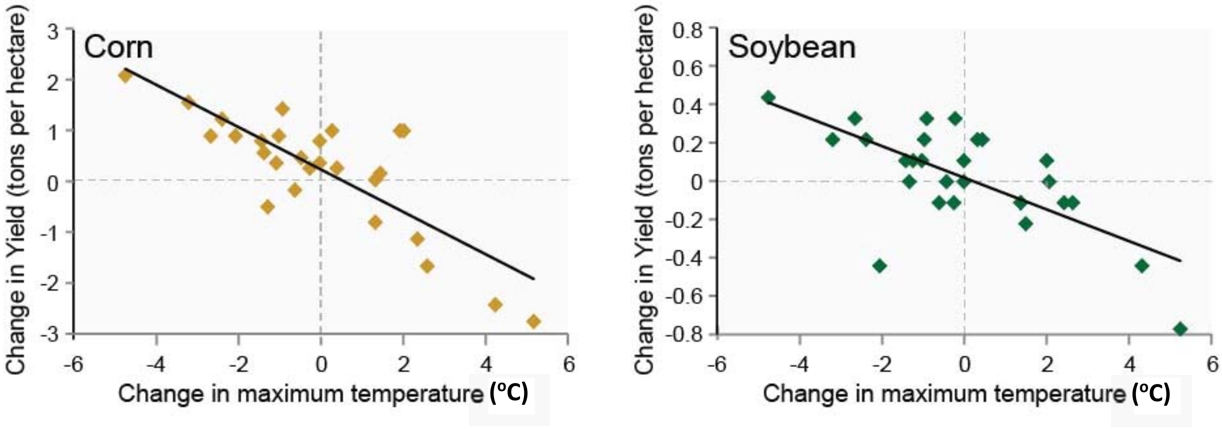

Fig. 8.3

(e) Describe and explain how increase in temperatures brought about by global warming impacts the crop yield of corn and soybean.

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1. **As the maximum temperature increases by up to 5 °C, the average crop yield for corn showed a corresponding decrease by 2 tons/hectare;**

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2. **and a similar trend was observed for the crop yield for soybean, where <u>average crop yield</u> <u>decreased</u> by <u>0.4 tons/hectare</u>;**

3. **This could be due to higher temperatures resulting in <u>sterile gametes/ sterile pollen grain/ low pollen grain production/ poor anther dehiscence/ less flowers form/ no flowers form/ low flower bud initiation</u>.;**

[Total: 10]

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9 A student carried out an investigation on yeast respiration using the following setup shown in Fig. 9.1. She put 5 grams of yeast into a glucose solution and put a layer of oil on top of the mixture. This layer of oil is impermeable to oxygen but not to carbon dioxide. The total volume of gas was recorded every 10 minutes for 1 hour and the results are shown in Table 9.1.

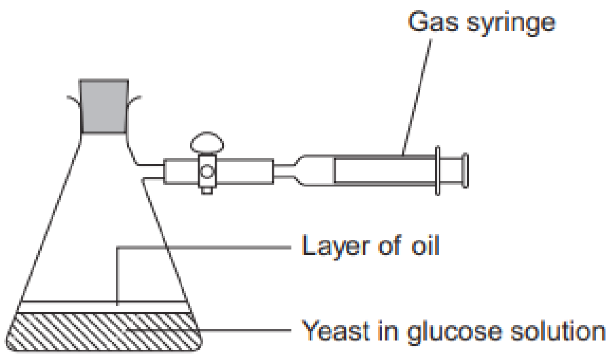

Fig. 9.1

Table 9.1

Time / minutes	Total volume of gas collected / cm^3
10	0.3
20	0.9
30	1.9
40	3.1
50	5.0
60	5.2

(a) **Ii)** Calculate the rate of gas production in $cm^3\ g^{-1}\ min^{-1}$ during the first 40 minutes of this investigation. Show your working clearly. [1]

Answer = ... $cm^3\ g^{-1}\ min^{-1}$

3.1 cm^3 / 5g / 40min = 0.0155 OR 0.016

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ii) Suggest why the rate of gas production decreased between 50 and 60 minutes.

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1. **Glucose becomes a limiting factor;**
   OR
2. **increase in ethanol / toxins build up causing yeast/cells to die;**
   (Note: ethanol disrupts yeast cell membrane causing cell death)

(b) The Respiratory Quotient (RQ) is defined as the ratio of carbon dioxide produced to oxygen consumed per unit time by an organism and the formula is as follows:

$$RQ = \frac{\text{volume of } CO_2 \text{ produced}}{\text{volume of } O_2 \text{ consumed}} \text{ per unit time}$$

Given that the rate of carbon dioxide produced equals to the rate of oxygen used for aerobic respiration, suggest and explain the RQ value for the setup shown in Fig. 9.1.

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1. RQ value **more than 1** / **higher** than RQ value for aerobic respiration;
2. **Oxygen** is **not consumed** during **anaerobic respiration** but **carbon dioxide** is **evolved** during **alcohol fermentation** by yeast;

(c) To show that the presence of oil has an effect on yeast respiration, the student also created another similar setup as the one shown in Fig. 9.1. All variables were kept constant in this second setup except that no oil was used.

Explain if there is any difference in the amount of ATP produced between both setups.

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1. **With oil: only 2 net ATP produced per glucose molecule, whereas without oil: 34/ 36/ 38 ATP produced per glucose molecule / about 17/ 18/ 19x more ATP produced for setup without oil as compared with setup with oil;**
2. **With oil, oxygen not present as the final electron acceptor (mark once), only glycolysis occurs to produce ATP;**
3. **without oil, oxygen is present as final electron acceptor (mark once), hence Krebs cycle and oxidative phosphorylation can occur to produce more ATP as well;**

In another experiment, the student investigated the effect of temperature on the rate of oxygen consumption of the lizard, *Sauromalus hispidus*. The body temperature of a lizard varies with environmental temperature.

Several baby lizards were fitted with small, airtight masks that covered their heads. Air was supplied inside the mask through one tube, and collected through another. The differences between oxygen concentrations in the air supplied for inhalation and the exhaled air enabled the researchers to measure the rate of oxygen consumption of the lizards.

The rate of oxygen consumption of each baby lizard was measured at different temperatures ranging from 15 °C to 40 °C and the average measurements were computed. The results are shown in Fig. 9.2.

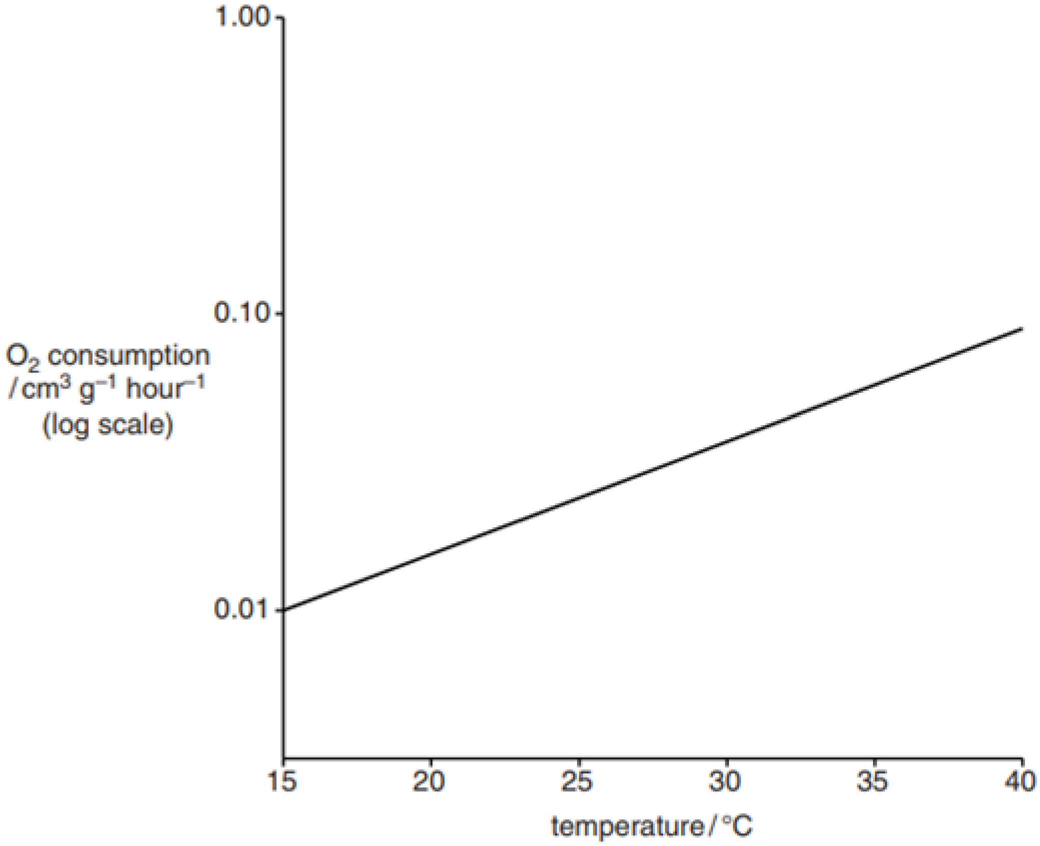

Fig. 9.2

(d) Using Fig. 9.2, suggest and explain how temperature influences the development of baby lizards.

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1. As temperature increase from 15 °C to 40 °C, the rate of oxygen consumption increased from 0.01 cm^{-3} g^{-1} $hour^{-1}$ to 0.09 cm^{-3} g^{-1} $hour^{-1}$;
2. because increase in temperature, increases kinetic energy of respiratory enzymes and substrates, leading to increased rate of effective collisions between enzyme and substrate / increased rate of formation of enzyme-substrate complexes;
3. Hence, with an increase in respiration rate, thus providing more ATP for growth, hence rate of development of baby lizards will increase;

[Total: 10]

10 The theory of continental drift was proposed over 90 years ago by Alfred Wegener and is now a widely accepted theory of how the continents were formed.

The supercontinent Gondwana is shown in Fig. 10.1. The land that was to form New Zealand began to break away from the supercontinent, Gondwana around 80 million years ago and broke away totally around 65 million years ago (Fig. 10.2). Ever since, New Zealand is now composed of North Island and South Island (Fig. 10.3).

The separation of New Zealand from Gondwana has led to some species like the New Zealand kiwi bird being distantly related to the rhea (South America), ostrich (Africa) and cassowary (Australia).

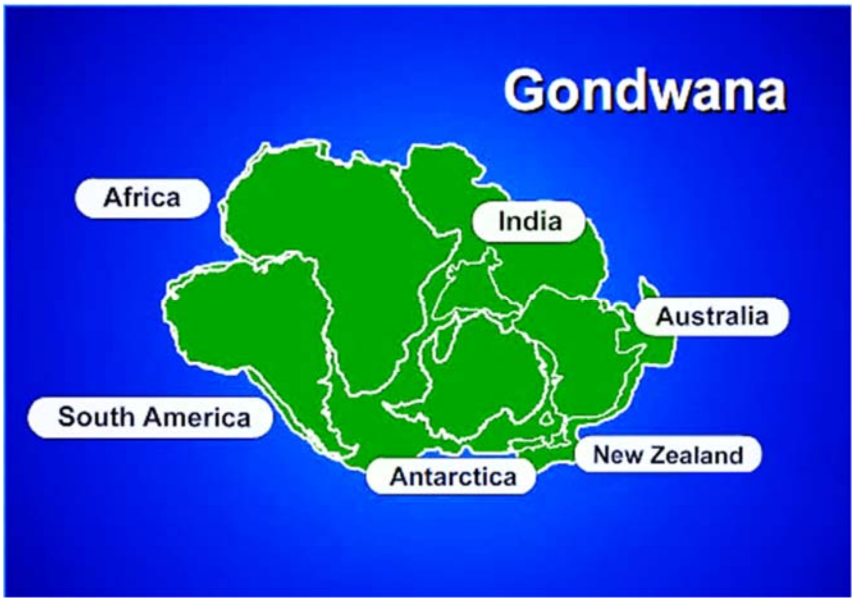

Fig. 10.1

Fig. 10.2

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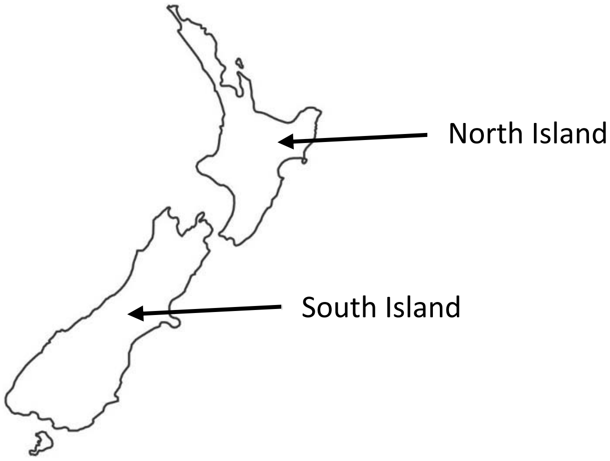

Fig. 10.3

(a) Using your knowledge of biogeography and continental drift, outline how the kiwi could be distantly related to some birds from other continents.

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1. **When <u>New Zealand separated from Gondwana</u>, it took a <u>subpopulation</u> of the <u>common ancestors</u> of the <u>kiwi bird, rhea, ostrich and cassowary</u> with it;**
2. **The subpopulation <u>evolved independently</u> to form the New Zealand kiwi birds.;**

For reasons that are not apparent, New Zealand was not inhabited by many mammals. Instead, its fauna became dominated by birds and insects. Birds became the predators, the scavengers, the herbivores and the insectivores. They lived everywhere from the highest mountains to the sea. Some have lost the ability to fly and became ground dwellers.

The kakapo, kea and kaka are parrots that are currently found in New Zealand (Fig. 10.4).

The kakapo (*Stringops habroptila*) is found only in New Zealand and is flightless. Highly endangered, they were relocated to several islands in New Zealand during the 1980s and 1990s.

The kea (*Nestor notabilis)* is the world's only alpine parrot, it can be found living near forested mountain habitats - Southern Alps on South Island. Kea are not found on North Island now.

Kaka (*Nestor meridionalis)* can be found on South Island and the sub-species of Kaka (*Nestor meridionalis septentrionalis*) is found on North Island. All Kaka live in the lowlands and mid-altitude regions.

kakapo kea kaka

Fig. 10.4

(b) Suggest a possible factor causing birds like the kakapo to become flightless early in their evolutionary history.

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1. **Lack of predators / mammals that hunted the kakapo for food;**
2. **AVP;**

Fig. 10.5 shows the phylogenetic relationships between the kakapo, kea and kaka derived from mtDNA evidence, together with the geological changes that shape the landscape of New Zealand.

Mya = Million years ago

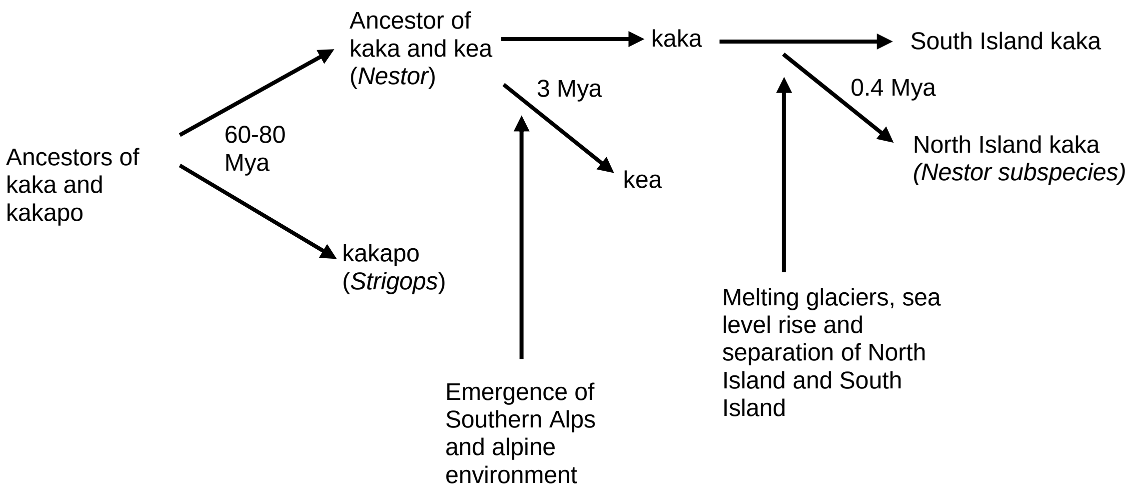

Fig. 10.5

(c) With reference to Fig. 10.5, explain the distribution and evolution of the kea.

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1. **Emergence of Southern Alps / alpine environment, 3 million years ago, provided opportunity for a founder population / ancestors of the kea to exploit this habitat / niche;**
2. **This sub-population was exposed to a different environment with different selection pressures compared to original population;**
3. **There is existing variation in this sub-population, those birds with favourable traits / adaptations for alpine environment have a selective advantage to the local conditions and will be selected for,**
4. **and will survive, reproduce and pass on their alleles to the next generation, thus increasing the frequency of favourable alleles;**
5. **Those best adapted to alpine environment remain in the alps, thus gene flow is disrupted.**
6. **As this sub-population evolved independently from other subpopulation, / accumulated different genetic mutations, allele frequencies changed due to genetic drift and natural selection;**
7. **Over hundreds and thousands of generations, the alpine population became reproductively isolated, no longer *interbreed* to produce *viable, fertile* offspring;**

(d) With reference to Fig. 10.5, suggest a possible reason for the emergence of a sub-species of kaka on North Island.

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1. **North and South Island become separated due to the land being covered by sea during melting of glaciers 0.4 million years ago;**
2. **Geographical isolation and independent evolution into a sub-species occurring in isolated North island kaka sub-population;**

OR

3. **Only a sub-species because 0.4 million years ago is insufficient time for complete (WTTE) speciation / reproductive isolation to take place;**
4. **Both North and South Island kaka can interbreed and produce fertile offspring if brought together;**

[Total: 10]